TP2/RR2 tests for American options on SPY (2025)

July 15, 2026

1 Summary results

The options data on SPY covers the period 1-Jan-25 to 29-Aug-25, for a total of 165 trading days.

Filters. Several conditions are applied in order to eliminate stale data.

1. **Volume filter.** Keep only options with non-zero trading volume.
2. **Non-zero bid filter.** Keep only options with non-zero bid price.
3. **Delta filter.** Keep only options with $|\Delta|$ in the range $[0.01, 0.99]$.

Table 1: The numbers of 4-uples in the data $n_{C,P}$, the violation counts $vio_{C,P}$ and the violation rates $vr_{C,P}$ for several data subsets.

Dataset	n_C	vio_C	vr_C	n_P	vio_P	vr_P
No Delta filter (mids)	67,115,004	3,823,583	5.697%	104,050,597	5,885,608	5.656%
Delta Filter (mids)	43,376,334	176,859	0.408%	62,305,648	1,182,236	1.897%
Delta Filter (bid-ask)	43,376,334	10,909	0.025%	62,305,648	232,228	0.373%

The violation counts and violation rates obtained for the entire dataset are shown in Table 1. The TP_2/RR_2 violations are computed using both mid-prices (second row) and using extreme bid-ask prices (third row).

For reference the first row shows the results obtained without applying the Delta filter, and mid prices. The Delta filter reduces the number of 4-uples and violations: its application leaves only 64.63% of the call 4-uples and only 59.88% of the put 4-uples. Furthermore, the Delta filter has an asymmetric impact on calls and puts, as it eliminates more call TP_2 violations than put RR_2 violations.

Daily violation rates. Figure 1 shows the daily violation rates, separately for calls and puts. The dashed lines show the average violation rates.

During the period considered, the puts violation rates dominates over that of the calls, except for a short period. This is consistent with the average violation rate which is much larger for puts than for calls.

Distribution in maturity. The distribution of the violations in maturities (T_1, T_2) is shown as heat maps in Figure 2 aggregating over the entire dataset.

The grey dots show all traded (T_1, T_2) pairs. The colored dots show points with violation rates above a certain threshold $(1\%, 2\%, 5\%)$.

The results show that violations for calls are larger near the diagonal $T_1 = T_2$, regardless of maturity. This is somewhat different from the SPX European options, where violations were concentrated at short maturities.

The RR_2 violations show also concentration in a band close to the diagonal, and also in a vertical band with $T_1 \sim 50 - 100$ days, regardless of T_2. Figure 3 shows the distribution of the largest RR2 violations, up to 15%, which show also a concentration in a band of $T_1 \in [50, 100]$ days.

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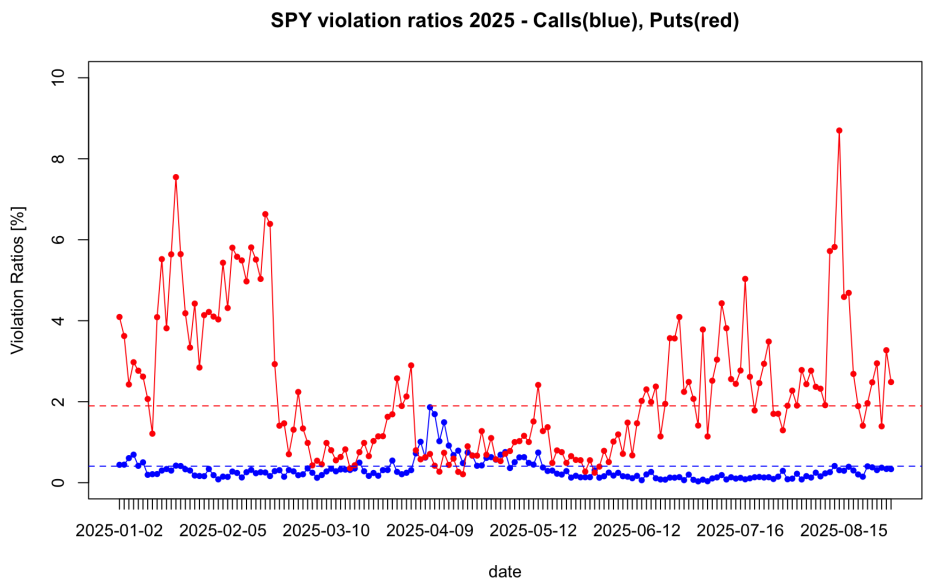


Figure 1: The violation rates for TP2/RR2 violations for each day in the 2025 dataset. The dashed lines show the average violation rates.

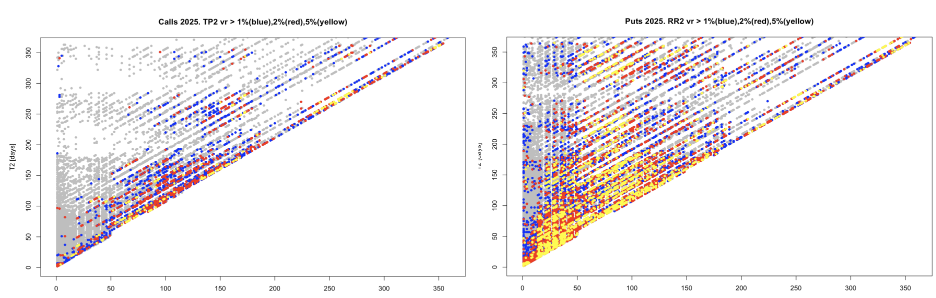


Figure 2: The distribution of TP2/RR2 violations in maturities (T_1, T_2), aggregating over all pricing days in the 2025 dataset for calls (left) and puts (right).

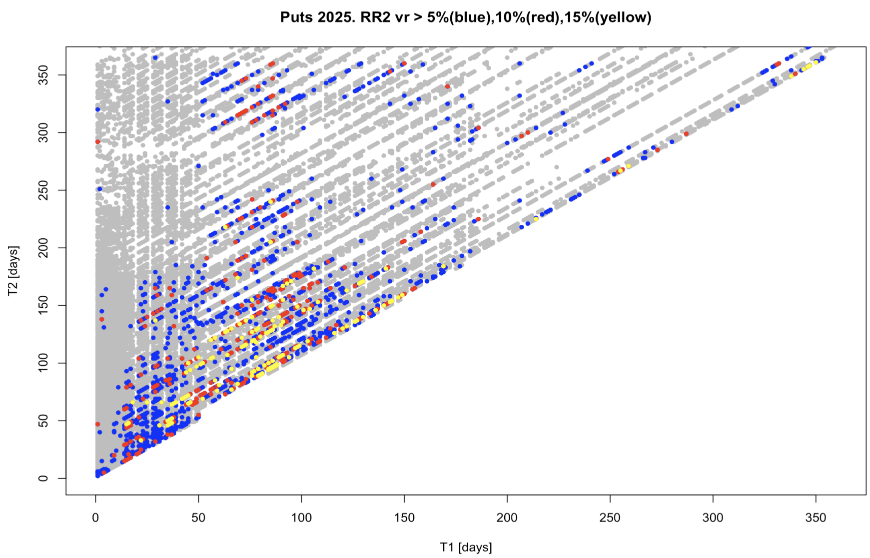


Figure 3: The distribution of largest RR2 violations in maturities (T_1, T_2), aggregating over all days in the 2025 dataset.

Distribution in Delta. For each TP2 violation we record $\Delta_{C1} := \Delta(C(K_1, T_1))$ and $\Delta_{C2} := \Delta(C(K_2, T_2))$. Same for puts, denoted as Δ_{P1}, Δ_{P2}.

Table 2: Mean and median of the distributions of the Delta for TP2/RR2 violating calls and puts.

	mean	median
Δ_{C1}	0.154	0.031
Δ_{C2}	0.147	0.035
Δ_{P1}	-0.045	-0.018
Δ_{P2}	-0.106	-0.075

The statistics of these four Deltas are shown in Table 2 and their density is shown in Figure 4. Most violations are for OTM options, although there is a small peak for deep ITM calls. This is similar to the upper island in Mike's Figure 5 (right). A similar peak is observed for ITM puts, although it is smaller. This is similar to the island in the upper right corner of Mike's Figure 7.

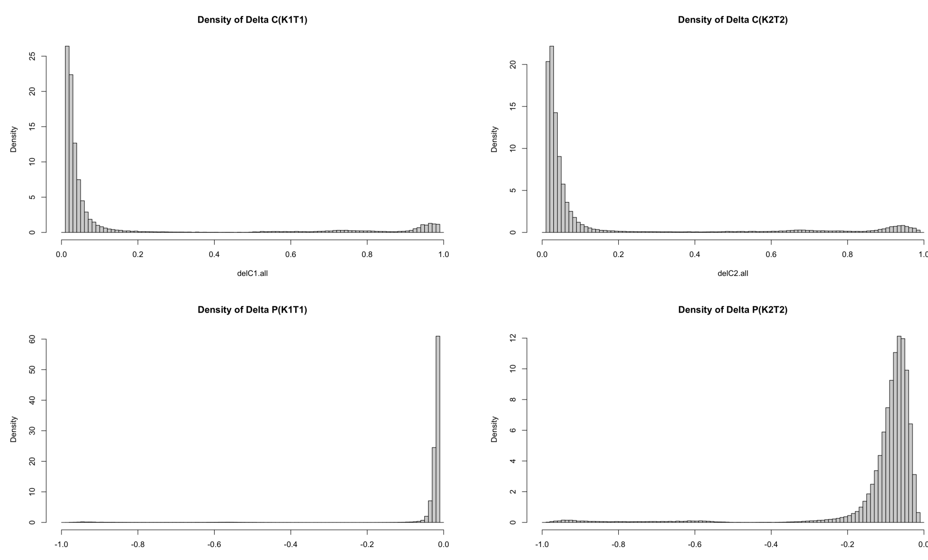
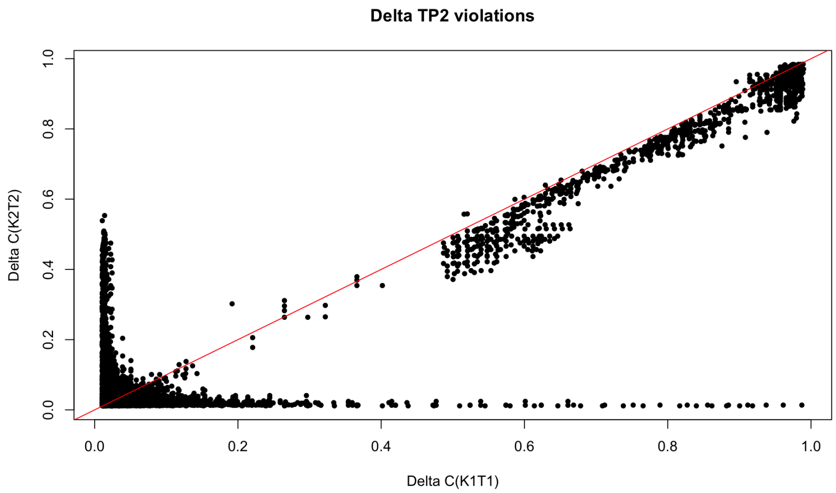


Figure 4: The probability density of the Deltas of $C(K_1T_1)$ and $C(K_2T_2)$ for TP2 violations (above) and of $P(K_1T_1)$ and $P(K_2T_2)$ for RR2 violations (below).

In order to see the correlation among $(\Delta_{C1}, \Delta_{C2})$ we show in Figure 5 a scatterplot with the two Deltas. These scatterplots show the concentration around the $(0, 0)$ corner (deep OTM options). Away from this point, many violations are concentrated near the diagonal. There is also the island in the right upper corner corresponding to deep ITM calls and lower left for puts.



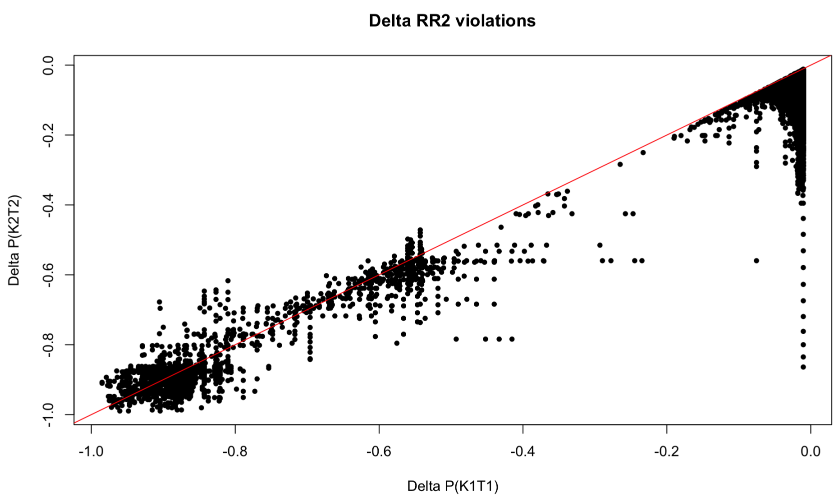


Figure 5: The joint distribution of $(\Delta_{C1}, \Delta_{C2})$ (above) and similar for puts (below). The red line shows the diagonal for reference.

Normalized violation ratio. Define a normalized violation ratio similar to that in Sec. 3.6 of Mike's note $R_c = \frac{c_{11}c_{22}-c_{12}c_{21}}{c_{11}c_{22}+c_{12}c_{21}}$ and analogous for puts.

The histogram of these ratios is shown in Figure 6. This shows $\log(count)$ for 1000 bins of the $[0, 1]$ interval. The ratios are very concentrated near zero. The median m, mean μ and the 95% quantile of these distributions are as follows.

$$\text{Calls: } m = 0.016, \mu = 0.031\,, Q95\% = 0.114 \tag{1}$$

$$\text{Puts: } m = 0.009, \mu = 0.012\,, Q95\% = 0.072 \tag{2}$$

Put violations are more frequent but are typically smaller.

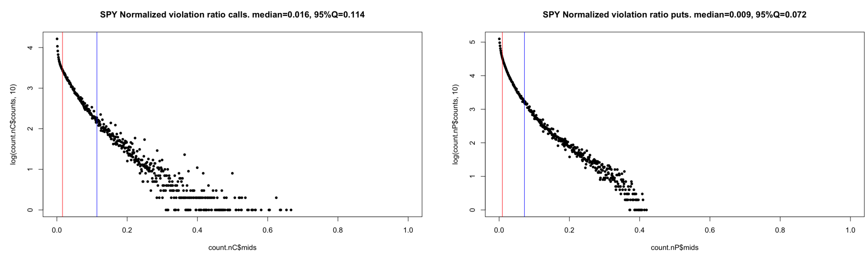


Figure 6: The count histograms of the normalized violation ratio for calls and puts.

2 Dividend-safe analysis

The impact of dividends can be eliminated by restricting the analysis to a safe dataset, for which no dividend is paid before T_1. Following the same approach as in Mike's note, I computed the violation rates for each day of Jan.25, Apr.25 and Jul.25 and restricting to option maturities less than 30 days.

The results are shown in Figure 7. The violation rates of the dividend-safe subset are generally larger than those of the entire data set (which includes all maturities, not only up to 30 days) but track them closely. Occasionally, as in the spikes observed in April 2025, the dividend-safe violation rates are much larger than those of the entire dataset.

The violation rates for the ex-dividend safe subset are close to those in Table 3 of Mike: for calls I get 0.826% (compared to 0.888%) and for puts I get 3.344% (compared to 2.895%). The violation rates for the full dataset are exactly the same as in Mike's Table 3.

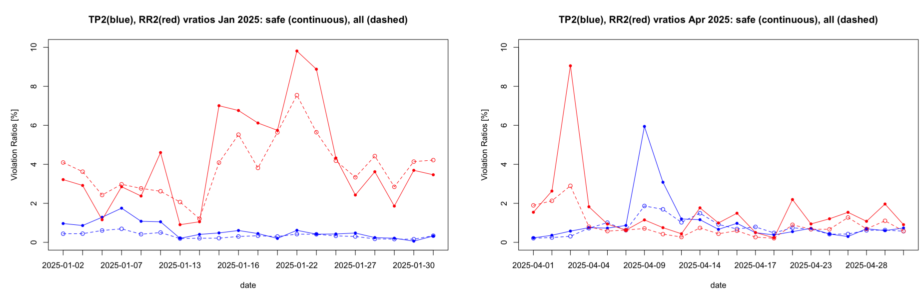
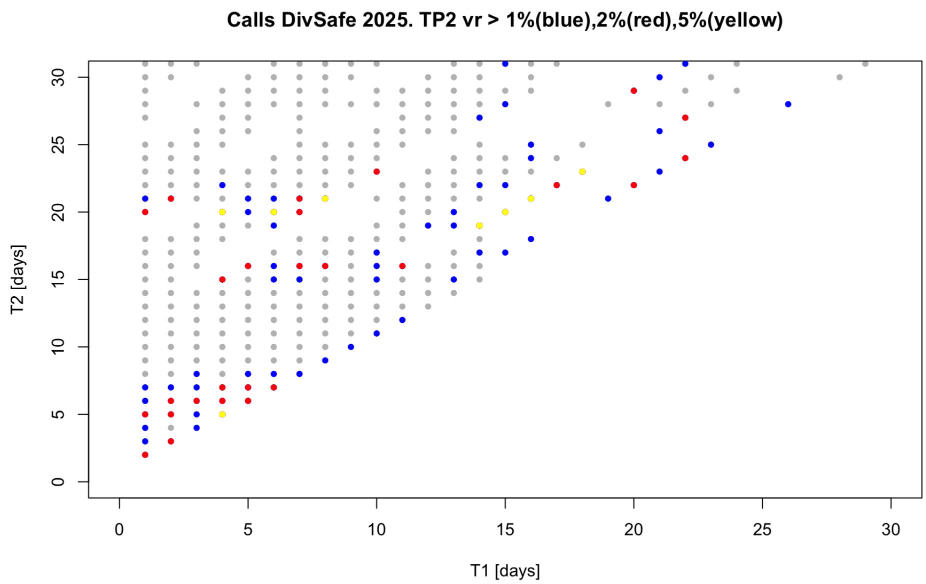


Figure 7: The violation rates for each day in the dividend-safe dataset (solid) and for all maturities (dashed) for Jan.25 (left) and Apr.25 (right).

Table 3: The violation rates for the safe subset and for the entire dataset.

Dataset	n_C	vio_C	vr_C	n_P	vio_P	vr_P
Jan 2025	835,704	4,827	0.578%	1,202,101	49,881	4.150%
April 2025	2,850,545	31,213	1.090%	3,038,414	57,742	1.430%
Jul 2025	1,041,896	3,038	0.292%	1,672,761	90,108	5.39%
Jan+Apr+Jul			0.826%			3.344%
All data	43,376,334	176,859	0.408%	62,305,648	1,182,236	1.897%

Distribution in maturities. The distribution in maturities of the violations for the dividend safe subset is shown in Figure 8, separately for calls and puts. The distribution confirms the 'diagonal effect' noted by Mike: largest violation rates are observed for close maturities $T_1 \sim T_2$.



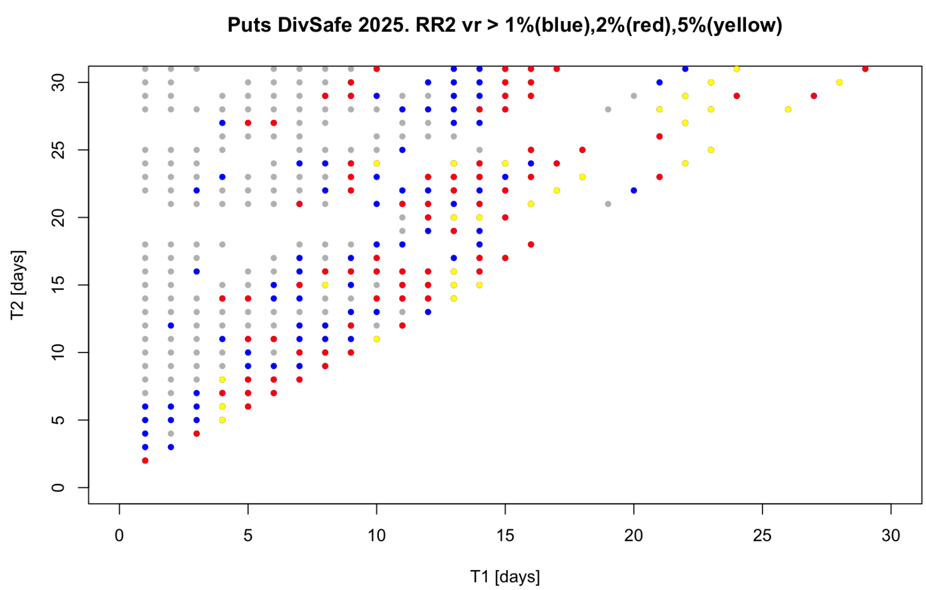


Figure 8: Heatmap for the violation rates in the dividend-safe dataset in maturities T_1, T_2.